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Hydroxyapatite nanowires rich in [Ca–O–P] sites for ethanol direct coupling showing high C_{6–12} alcohol yield†

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Herein, we have shown that the [Ca–O–P] sites exposed on hydroxyapatite are clearly responsible for C–C formation in ethanol direct-coupling, and their high density accelerates the C–C coupling rate and boosts C_{6–12} alcohol production. Notably, nanowire-like hydroxyapatite exhibited 30.4% selectivity to *n*-butanol and 63.9% selectivity to C_{6–12}OH at a conversion of 45.7% at 325 °C, and thereby close to 30% yield of C_{6–12}OH, which is greatly higher than that using the state-of-the-art catalysts (6%).

The increased availability of ethanol from biomass¹ drives the interest of both academic and industrial researchers to upgrade this platform molecule into more valuable chemicals, instead of the energy intensive and environmentally concerning petroleum processes.² Ethanol can undergo C–C formation to yield value-added *n*-butanol and C_{6–12} aliphatic alcohols.³ In industry, *n*-butanol and C_{6–12} aliphatic alcohols are usually produced *via* the hydroformylation of α -alkenes followed by the hydrogenation of the resulting aldehydes.⁴ These long carbon chain products are desirable as biofuels or additives due to their high energy density (29 MJ L^{−1} for *n*-butanol and 32–35 MJ L^{−1} for C_{6–12} aliphatic alcohols) and low miscibility with water compared with ethanol.⁵ In the past years, several active catalysts such as hydroxyapatite,⁶ Mg–Al mixed oxides,⁷ and Cu–CeO₂/carbon⁸ have been reported for ethanol coupling reactions. Among them, hydroxyapatite [HAP, Ca₁₀(OH)₂(PO₄)₆] is particularly noteworthy owing to its good activity and selectivity predominantly for *n*-butanol (~70%).⁹ Although great efforts were put into this field, selective production of C_{6–12} aliphatic alcohols from ethanol still remains an unsolved challenge.¹⁰ The main reason for this is that the C_{6–12} aliphatic alcohols are usually obtained as side products following the formation of

n-butanol, owing to the uncontrolled cascade-type self- and cross-coupling reactions.¹¹ In this study, we have succeeded in the preparation of a nanowire-like HAP catalyst rich in [Ca–O–P] sites which increases the coverage of alcohol reactants, thus accelerating the C–C coupling rate and boosting C_{6–12} alcohol production. The C–C formation was identified *via* the ethanol direct-coupling route, rather than the Guerbet path, based on catalytic experiments and *in situ* FTIR studies.

Two types of HAP catalysts (nanowires and nanofibers) with different densities of [Ca–O–P] sites were prepared, for comparison, by the modified solution chemistry method (ESI†), where CaCl₂ and NaH₂PO₄ were used as reactants and oleate was adopted as a structure-directing agent. As a control, HAP nanorods were prepared *via* a conventional method using Ca(NO₃)₂ and (NH₄)₂HPO₄ as reactants without surfactants.^{9a} XRD analysis (Fig. 1a) showed that all the as-prepared HAP catalysts present typical characteristics of crystalline apatite with a hexagonal phase structure. The great difference of the (300)/(002) intensity ratio for these HAP catalysts is

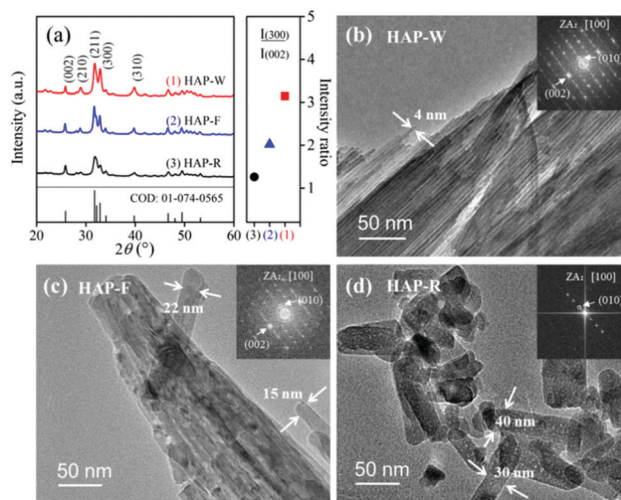


Fig. 1 (a) XRD patterns. TEM images of HAP-W (b), HAP-F (c), and HAP-R (d).

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an indication of their aspect ratios.¹² TEM images in Fig. 1b–d exhibited these samples showing different diameters, *i.e.*, ~4, 15–22, and 30–40 nm, in sequence, although there exists particle stacking. Their aspect ratios are correspondingly >400, 80–200, and 2–5, based on TEM observation (Fig. S1, ESI†). Therefore, we defined them as HAP nanowires (HAP-W, Fig. 1b), HAP nanofibers (HAP-F, Fig. 1c), and HAP nanorods (HAP-R, Fig. 1d). The surface area is 17, 20, and 32 m² g⁻¹ for HAP-W, HAP-F, and HAP-R, respectively (Table S1, ESI†). FTT of HRTEM images (Fig. 1 and Fig. S2, ESI†) over these HAP catalysts viewed along the [001] direction (*c*-axis) reveals that the most frequently exposed lateral termination facet is the (010) plane, where both Ca²⁺ and PO₄³⁻ are exposed,^{6a,13} which agrees well with other reports that HAP tends to grow along the *c*-axis during wet chemistry synthesis.^{12,13}

The ethanol coupling activity on HAP-W is compared to those on HAP-F and HAP-R by examining the selectivity of *n*-butanol and C_{6–12} alcohols (C_{6–12}OH) as a function of ethanol conversion (Fig. 2a and Table S2, ESI†). One can see that as the conversion increases from 5 to 47%, the selectivity of C_{6–12}OH (including different isomers, Fig. S3, ESI†) increases at the expense of *n*-butanol. However, product distributions over these catalysts are greatly different at the same ethanol conversion. The selectivity towards C_{6–12}OH on HAP-W is 8% higher than that on HAP-F, and is 17% higher than that on HAP-R. In contrast, the selectivity of *n*-butanol shows an opposite trend, consistent with the literature.¹¹ The optimized HAP-W exhibits 30.4% selectivity to *n*-butanol and 63.9% selectivity to C_{6–12}OH in a 45.7% conversion. To shed light on the outstanding features of HAP-W, the yields of higher alcohols and the ones in the literature are shown in Fig. 2b as a function of the reaction rate. The high yield (29.3%) of C_{6–12}OH, which is greatly higher than that using the state-of-the-art catalysts (*ca.* 6%), at an identical reaction rate (*ca.* 8.0 mmol g_{cat}⁻¹ h⁻¹) shows that HAP-W is competent for the efficient production of C_{6–12}OH from ethanol.

To clarify the difference in product distributions, the surface basicity and acidity of these HAP catalysts were evaluated, since the present acid–base pairs catalyze the deprotonation of ethanol or its derivatives to form carbanions and the subsequent C–C formation.^{3b} The surface basicity was firstly measured by the CO₂ temperature-programmed desorption mass spectrometry (CO₂-TPD-MS) experiments. As shown in Fig. 3a,

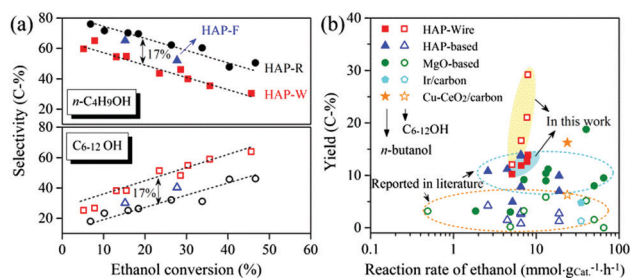


Fig. 2 (a) Dependence of product selectivity on the conversion varied by changing the WHSV. Reaction conditions: 325 °C, C₂H₅OH/N₂ = 5.7 : 94.3, N₂ 28 mL min⁻¹, WHSV = 0.8–5 g_{C₂H₅OH} g_{cat}⁻¹ h⁻¹. See Table S2 (ESI†) for detailed information. (b) Yields of *n*-butanol and C_{6–12}OH versus ethanol reaction rate for catalysts described in this work and in the literature (2–18, Table S3, ESI†).

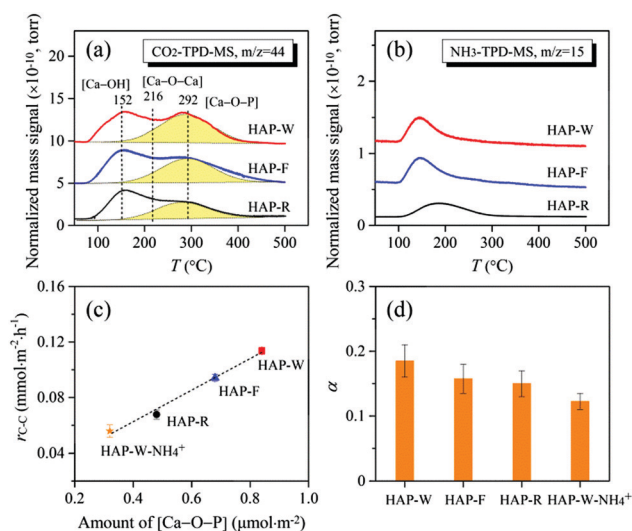


Fig. 3 (a) CO₂- and (b) NH₃-TPD-MS profiles. (c) Correlation of the formation rate of C–C bonds with the density of [Ca–O–P] sites (normalized to the specific surface area); (d) α number of HAP catalysts. The suffix “–NH₄⁺” indicates that HAP-W was treated with NH₄NO₃ aqueous solution at 25 °C for 2 h.

three CO₂ desorption peaks are deconvoluted on all these samples at ~152, 216, and 292 °C, which correspond to CO₂ desorbing from –OH bound to Ca²⁺ (weak, denoted as [Ca–OH]), O²⁻ bound to Ca²⁺ (medium, denoted as [Ca–O–Ca]), and O²⁻ of PO₄³⁻ bound to Ca²⁺ (*i.e.*, CaO/PO₄³⁻ pairs, strong, denoted as [Ca–O–P]), respectively, on the basis of CO₂-IR analysis.^{6a,14} As expected, the normalized density of the strong basic sites increases from 0.48 μmol m⁻² on HAP-R, to 0.68 μmol m⁻² on HAP-F, and then, to 0.84 μmol m⁻² on HAP-W (Table S1, ESI†). This finding is attributed to the anisotropic characteristic of HAP.¹³ The *a*-surface (*i.e.*, (010) and (100)) is rich in phosphate and hydroxide ions, suggesting the basic feature. In contrast, the basal *c*-surface (*i.e.*, (001) plane) is rich in Ca²⁺ sites. Therefore, the higher aspect ratio of HAP is, the more frequent (010) facets expose and hence the higher density of [Ca–O–P] sites; while, the intensities of CO₂ signals from [Ca–OH] and [Ca–O–Ca] sites are similar. Overall, HAP-W exhibits more [Ca–O–P] sites compared with the two others. Furthermore, NH₃-TPD studies were carried out to evaluate the acidity (Fig. 3b). In this case, all HAP catalysts display a broad desorption peak in the range of 100–300 °C, suggesting that the weak-medium acid sites (Ca²⁺ and P–OH sites) are present, which agrees well with the reported probe-IR and -XPS analysis.^{14a,15} A slightly different acid strength between HAP-F/W and HAP-R may correspond to different exposure densities of Ca²⁺ and/or P–OH sites on the surface. Specifically, the P–OH sites may show a slightly stronger acidity as compared to Ca²⁺. CO₂- and NH₃-TPD analysis proved that the HAP surface presents both acid and base sites, which are crucial for C–C formation,^{6a} and only the density of strong base sites obviously changed.

Next, the C–C bond formation rate was calculated and plotted as a function of the [Ca–O–P], [Ca–OH] or acid site density (Fig. 3c). The rate was calculated at ~16% conversion and a WHSV of ~2 g_{C₂H₅OH} g_{cat}⁻¹ h⁻¹ on three HAP catalysts, which was normalized to their specific surface area respectively. As shown

in Fig. 3c and Fig. S4 (ESI[†]), a linear relationship was only found between the C–C bond formation rate and the [Ca–O–P] density, but not for the others, indicating that the strong basic [Ca–O–P] site is responsible for C–C formation, in accord with the literature.^{6a} The density of [Ca–O–Ca] is one order of magnitude lower than that of [Ca–O–P] or [Ca–OH], and no linear relationship is found between [Ca–O–Ca] and the reactivity. In addition, the presence of POH and (Ca)–OH acid–base pairs can't be excluded, but no clear relationship was found between these sites and the reactivity.^{15b} This finding means that the self- and cross-coupling of *n*-butanol with ethanol is promoted on HAP-W compared to that on HAP-R and HAP-F, thus boosting the formation of C_{6–12} alcohols (Fig. 2a).

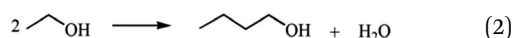
To shed more light on the role of [Ca–O–P], NH₄NO₃ solution was used to selectively remove these basic sites by substituting NH₄⁺ for Ca²⁺, denoted as HAP-W-NH₄⁺.¹⁶ No structural change was observed based on XRD data (Fig. 1a and Fig. S5, ESI[†]). The density of the [Ca–O–P] sites on HAP-W-NH₄⁺ diminishes by ~60% while those of the others (*i.e.*, [Ca–OH] and [Ca–O–Ca]) remain unchanged (Fig. S6 and Table S1, ESI[†]). As seen in Fig. 3c, the C–C formation rate decreases from 0.114 mmol m^{–2} h^{–1} of HAP-W to 0.056 mmol m^{–2} h^{–1} of HAP-W-NH₄⁺.

Since the reaction rate constants (*k*) are similar for self- and cross-coupling reactions between ethanol and its condensation products,¹¹ the rate of product formation depends mainly on the surface coverage of the reactants ($r = k[A]^m[B]^n$, *m* and *n*: reaction order, *m* = *n* and *A* = *B* for self-coupling reactions). The fraction of oxygen-containing groups (α)¹¹ consumed during ethanol coupling is available to be a descriptor for quantitatively evaluating the surface coverage of reactants on HAP. This α can be calculated using a “pseudo step-growth” model:¹¹

$$W_n/W_0 = n \times (1 - \alpha)^2 \times \alpha^{(n-2)/2} \quad (1)$$

where *W_n*, *W₀*, and *n* are the mass of *n* (*n* ≥ 4) species, the mass of ethanol, and the carbon number of aliphatic alcohols, respectively. As expected, HAP-W gives a larger α than other HAP catalysts, *i.e.*, 0.19 for HAP-W, 0.16 for HAP-F, 0.14 for HAP-R, and 0.12 for HAP-W-NH₄⁺ (Fig. 3d).

For ethanol coupling on HAP, the formation of *n*-butanol involves the first C–C formation. Thus, unraveling the formation route of *n*-butanol would help us to understand the total reaction network. The coupling of ethanol to *n*-butanol has been proposed to progress *via* at least two different routes: “direct-coupling”¹⁷ and “Guerbet”.^{6a,9,11,18} The direct-coupling route involves direct coupling of two ethanol molecules with apparently no intermediates (eqn (2)):



In contrast, when the Guerbet reaction dominates the coupling process, ethanol firstly dehydrogenates to H₂ and acetaldehyde, which undergoes self-condensation, dehydration, and hydrogenation to yield *n*-butanol (eqn (3)–(5)):

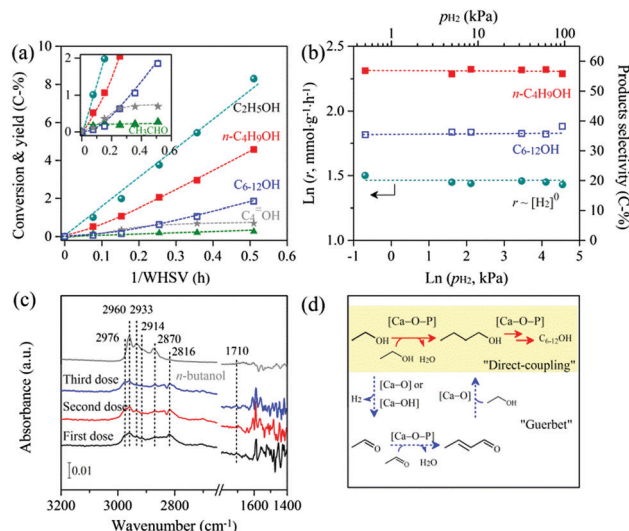
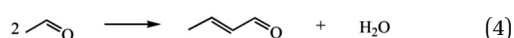


Fig. 4 (a) Dependence of the product distributions on the residence time (1/WHSV). (b) Dependence of ethanol reaction rates and product selectivity on H₂ pressure. (c) *In situ* IR spectra collected right after each dose of ethanol (5 × 10^{–8} mol). (d) Proposed reaction routes for aliphatic alcohol formation. Experiments were tested at 325 °C for (a, b) and 300 °C for (c).



To reveal the possible coupling route of ethanol, the effect of residue time on product distributions over HAP-W was firstly studied in order to identify primary and secondary reactions. Ethanol conversion and product yield were presented as a function of residue time (Fig. 4a). Local slopes of the curves indicate the formation rate of products at a specific conversion and residence time. The high initial slope for *n*-butanol shows that *n*-butanol is a main primary product. The initial slope of C_{6–12}OH is near to zero, indicating that they form in secondary reactions *via* the self- and cross-coupling of *n*-butanol and ethanol. A small amount of acetaldehyde/2-butenol and no 2-butenol are observed at 1/WHSV < 0.5 h. The presence of acetaldehyde suggests that a small amount of ethanol undergoes dehydrogenation. These results support that the *n*-butanol forms *via* at least two parallel routes: direct coupling of ethanol and sequential paths involving the hydrogenation of 2-butenol derived from acetaldehyde (Guerbet reaction). The steep initial slope (inset panel) suggests that ethanol direct-coupling is the dominant path for the formation of aliphatic alcohols. In this case, ethanol reaction rates would be independent of H₂ pressure because H₂O is the only coproduct (eqn (2)). However, when the Guerbet reaction was dominant, H₂ would play a detrimental role, because of high H₂ pressure hindering the dehydrogenation of ethanol (eqn (3)–(5)). Therefore, a series of experiments were conducted by cofeeding H₂–ethanol over HAP-W. Fig. 4b shows that the ethanol reaction rate was almost constant when the H₂ pressure increased from 0.5 to 94 vol% (*i.e.*, $r \sim [\text{H}_2]^0$). Furthermore, the selectivity of *n*-butanol, C_{6–12}OH, 2-butenol, and acetaldehyde remained unchanged (Fig. 4b and Fig. S7, ESI[†]). When only ethanol was fed, very small amounts of H₂ were detected (Table S2, ESI[†]). In addition, no deuterium atoms

incorporated into *n*-butanol produced over HAP using D₂ as the carrier gas,¹⁸ which precludes the possibility that H species derived from the ethanol dehydrogenation, if possible, interact with the HAP surface and then incorporate into *n*-butanol. Scalbert *et al.*¹⁷ also reported quantitative evidence of the majority role of the “direct coupling” over the route involving acetaldehyde on HAP, and they proposed that intermediate hydrogenation involves H-transfer from an ethanol molecule (instead of H₂).

Next, observation of surface species aldehydes on HAP-W was conducted by *in situ* IR (Fig. 4c). In the first 1 min, pronounced peaks are observed in the $\nu(\text{C-H})$ region. The peaks centered at 2976 and 2914 cm⁻¹ are assigned to $\nu_{\text{as}}(\text{CH}_3)$ and $\nu_{\text{as}}(\text{CH}_2)$ of ethanol, respectively.¹⁹ Ethanol dissociated on acid-base sites will elongate C-O bonds, thus shifting $\nu_{\text{as}}(\text{CH}_2)$ to a lower wavenumber, ~ 2820 cm⁻¹.²⁰ A peak at 2960 cm⁻¹, together with bands at 2933 and 2870 cm⁻¹, corresponds to *n*-butanol compared with the reference spectrum (the gray line). The coexistence of ethanol and *n*-butanol right after one dose of ethanol demonstrates the fast coupling of ethanol. Importantly, no signal at ~ 1710 cm⁻¹ typically assigned to the $\nu(\text{C=O})$ mode of acetaldehyde is detected,²¹ even though after the second and third doses. This observation agrees with the finding reported by Hanspal *et al.* that the active sites responsible for acetaldehyde formation are much fewer than that for *n*-butanol.^{19b} Furthermore, in our recent work,^{3c} cobalt-HAP catalysts were used to directly produce aromatic oxygenates from ethanol, which is initiated by acetaldehyde derived from ethanol *via* dehydrogenation on cobalt. In that case, the signal at 1710 cm⁻¹ of acetaldehyde was discernibly observed by *in situ* IR, even though the ethanol feeding (5×10^{-8} mol) was very low, demonstrating that the absence of acetaldehyde in this work was not because of the unsatisfactory time resolution.

Therefore, the above results objectively prove that the upgrading of ethanol to aliphatic alcohols on HAP-W proceeds mainly *via* the “direct-coupling” route (Fig. 4d, solid line).¹⁷ Meanwhile, the Guerbet route also contributes a small fraction to aliphatic alcohol formation (Fig. 4d, dashed line).^{6a,9b,11} Differently, several previous studies concluded the upgrading of ethanol to aliphatic alcohols *via* the Guerbet route,^{6,15b} whose kinetic and IR studies were performed usually after a long time (hours) on stream. In fact, the active sites might change from [Ca-O-P] to [Ca-OH] and [P-OH] because of hydrolysis after the long reaction time (Fig. S8, ESI†).^{11,22} These strong basic sites can be regenerated after high temperature treatment (Table S4, ESI†). However, in this work, IR studies were performed right away after starting the reaction using fresh catalyst.

In summary, 30.4% *n*-butanol selectivity and 63.9% C₆₋₁₂OH selectivity were achieved in the ethanol coupling reaction on wire-like HAP at a conversion of 45.7% at 325 °C, thereby showing nearly 30% yield of C₆₋₁₂OH. This yield is greatly higher than that using the state-of-the-art catalysts (6%). The [Ca-O-P] site is responsible for C-C formation and its high density increases the coverage of alcohol reactants, thus accelerating the C-C coupling rate and boosting C₆₋₁₂ alcohol production. Ethanol coupling on wire-like HAP proceeds mainly *via* the “direct-coupling” route to form aliphatic alcohols.

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Conflicts of interest

There are no conflicts to declare.

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